Abstract Algebra I

Lecture notes, Chapters 1–5 (Sections 0–28)

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Chapter 1. Groups and Subgroups

These notes follow Fraleigh, *A First Course in Abstract Algebra*; the book's numbering is given in brackets.

1.1 Sets and relations (§0)

Motivation. Algebra studies sets equipped with operations, and almost every construction in these notes (cosets, factor groups, residue classes, fields of quotients) is a set of equivalence classes. Theorem 0.22 is the reason such constructions are legitimate: an equivalence relation is the same thing as a partition, so "identifying" equivalent elements always produces a well-defined new set.

A **relation** between sets A and B is a subset $\mathcal{R} \subset A \times B$; a **function** $\phi : A \to B$ is a relation in which each $a \in A$ appears as first member of exactly one pair (a, b), written $b = \phi(a)$. ϕ is **one-to-one** (injective) if $\phi(a_1) = \phi(a_2)$ implies $a_1 = a_2$, **onto** (surjective) if every b is a value, and a **bijection** if both. Sets have the same **cardinality** if there is a bijection between them. A **partition** of S is a collection of nonempty disjoint subsets (**cells**) whose union is S. A relation $\sim$ on S is an **equivalence relation** if it is reflexive ($x \sim x$), symmetric ($x \sim y \Rightarrow y \sim x$) and transitive ($x \sim y$, $y \sim z \Rightarrow x \sim z$).

Theorem 1.1 (0.22, equivalence relations and partitions)

Let $\sim$ be an equivalence relation on a nonempty set S. The sets $\bar{a} = \{x \in S : x \sim a\}$ form a partition of S. Conversely, each partition of S gives the equivalence relation "$x \sim y$ iff x, y lie in the same cell".

Proof. Each a lies in $\bar{a}$ (reflexivity), so the cells are nonempty and cover S. Suppose $c \in \bar{a} \cap \bar{b}$. If $x \in \bar{a}$, then $x \sim a$, $a \sim c$ (symmetry) and $c \sim b$, so $x \sim b$ (transitivity) and $\bar{a} \subset \bar{b}$; symmetrically $\bar{b} \subset \bar{a}$. So distinct cells are disjoint.

Conversely, every x is in a cell with itself; if x, y share a cell so do y, x; and if x, y share a cell and y, z share a cell, these cells contain y and hence coincide. ■

Example 1.1

Congruence modulo n: for $n \in \mathbb{Z}^+$, $h \equiv k \pmod{n}$ iff n divides $h - k$. It is an equivalence relation whose cells are the n **residue classes modulo** n.

1.2 Binary operations and isomorphisms (§§1–3)

Motivation. $\langle \mathbb{R}, + \rangle$ and $\langle \mathbb{R}^+, \cdot \rangle$ look different but behave identically once we match $x \leftrightarrow e^x$. Isomorphism makes "behaves identically" precise; from now on we care only about properties that survive isomorphisms, and the way to prove two structures different is to find such a structural property held by one but not the other.

A **binary operation** $*$ on S is a function $S \times S \to S$. It is **commutative** if $a * b = b * a$ and **associative** if $(a * b) * c = a * (b * c)$ for all a, b, c. A subset H is **closed** under $*$ if $a * b \in H$ for $a, b \in H$. Examples: $+$ and $\cdot$ on $\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$; on the circle $U = \{z \in \mathbb{C} : |z| = 1\}$, multiplication; on $\mathbb{R}_c = [0, c)$, addition modulo c; on $U_n = \{z : z^n = 1\}$ and $\mathbb{Z}_n = \{0, 1, \ldots, n-1\}$, multiplication and addition modulo n.

Theorem 1.2 (2.13, associativity of composition)

For functions $f, g, h : S \to S$, $f \circ (g \circ h) = (f \circ g) \circ h$.

Proof. For $x \in S$, both sides send x to $f(g(h(x)))$. ■

A **binary algebraic structure** $\langle S, * \rangle$ is a set with a binary operation. An **isomorphism** $\phi : \langle S, * \rangle \to \langle S', *' \rangle$ is a bijection with $\phi(x * y) = \phi(x) *' \phi(y)$. To show two structures are isomorphic: define ϕ, show it is one-to-one and onto, and verify the homomorphism property. To show they are not, find a **structural property** (preserved by every isomorphism, such as cardinality, commutativity, or existence of solutions of $x * x = c$) held by one and not the other.

Example 1.2

$\phi(x) = e^x$ is an isomorphism $\langle \mathbb{R}, + \rangle \to \langle \mathbb{R}^+, \cdot \rangle$. $\langle \mathbb{R}_{2\pi}, +_{2\pi} \rangle \cong \langle U, \cdot \rangle$ via $x \mapsto e^{ix}$, and $\langle \mathbb{Z}_n, +_n \rangle \cong \langle U_n, \cdot \rangle$ via $k \mapsto e^{2\pi i k/n}$. $\langle \mathbb{Q}, + \rangle \ncong \langle \mathbb{R}, + \rangle$ (cardinality), $\langle \mathbb{Z}, + \rangle \ncong \langle \mathbb{Z}^+, + \rangle$ (in $\mathbb{Z}$, $x + x = c$ has a solution for $c = 0$), and $\langle \mathbb{R}, \cdot \rangle \ncong \langle \mathbb{C}, \cdot \rangle$ ($x \cdot x = c$ is solvable for every $c \in \mathbb{C}$).

An **identity element** e satisfies $e * s = s * e = s$ for all s; it is unique ($e = e * e' = e'$).

Theorem 1.3 (3.14)

If $\langle S, * \rangle$ has an identity e and $\phi : \langle S, * \rangle \to \langle S', *' \rangle$ is an isomorphism, then $\phi(e)$ is an identity for $*'$.

Proof. For $s' \in S'$ write $s' = \phi(s)$ (onto). Then $\phi(e) *' s' = \phi(e * s) = \phi(s) = s'$, and similarly $s' *' \phi(e) = s'$. ■

1.3 Groups (§4)

Motivation. Solving $a * x = b$ needs exactly three things: regrouping (associativity), a neutral element, and the ability to undo a (an inverse). A group is the abstraction of "an operation in which every equation $a * x = b$ can be solved uniquely"; numbers under addition, nonzero numbers under multiplication, and symmetries of an object all share this.

Definition (Group)

A **group** $\langle G, * \rangle$ is a set with a binary operation such that
$\mathcal{G}_1$: $*$ is associative;
$\mathcal{G}_2$: there is $e \in G$ with $e * x = x * e = x$ for all x;
$\mathcal{G}_3$: each a has an inverse a' with $a' * a = a * a' = e$.
It is **abelian** if $*$ is commutative.

Theorem 1.4 (4.15, cancellation laws)

In a group, $a * b = a * c$ implies $b = c$, and $b * a = c * a$ implies $b = c$.

Proof. Multiply $a * b = a * c$ on the left by a' and use associativity: $b = e * b = (a' * a) * b = a' * (a * b) = a' * (a * c) = c$. The right law is symmetric. ■

Theorem 1.5 (4.16)

In a group, the equations $a * x = b$ and $y * a = b$ have unique solutions x, y.

Proof. $x = a' * b$ is a solution, and any two solutions satisfy $a * x_1 = a * x_2$, hence are equal by Theorem 1.4. Similarly $y = b * a'$. ■

Theorem 1.6 (4.17)

In a group the identity element is unique, and each element has a unique inverse.

Proof. Identity: $e = e * e' = e'$. If a' and a'' are inverses of a, then $a * a' = e = a * a''$, so $a' = a''$ by cancellation. ■

We write a^{-1} for the inverse and, multiplicatively, a^n for powers ($a^0 = e$, $a^{-n} = (a^{-1})^n$); additively, $-a$ and na.

Corollary 1.7 (4.18)

$(a * b)' = b' * a'$.

Proof. $(a * b) * (b' * a') = a * (b * b') * a' = a * a' = e$, so $b' * a'$ is the inverse of $a * b$ by uniqueness. ■

Example 1.3

$\langle \mathbb{Z}, + \rangle$, $\langle \mathbb{Q}^*, \cdot \rangle$, $\langle \mathbb{Z}_n, +_n \rangle$, $\langle U_n, \cdot \rangle$, and $GL(n, \mathbb{R})$ (invertible $n \times n$ matrices under multiplication, nonabelian for $n \geq 2$) are groups; $\langle \mathbb{Z}^+, + \rangle$ and $\langle \mathbb{Z}, \cdot \rangle$ are not. Up to isomorphism there is exactly one group of each order $1, 2, 3$ (the group table is forced: each element appears exactly once in each row and column, by Theorem 1.5), and two groups of order 4: $\mathbb{Z}_4$ and the Klein 4-group V, in which every element is its own inverse.

1.4 Subgroups (§5)

Motivation. Just as one studies a vector space through its subspaces, a group is understood through the smaller groups sitting inside it. The two results here give a quick test for being a subgroup and the most important source of subgroups: the powers of a single element.

$H \subset G$ is a **subgroup**, $H \leq G$, if it is closed under the operation of G and is a group under it. The **order** $|G|$ of G is its number of elements.

Theorem 1.8 (5.14)

$H \subset G$ is a subgroup iff (1) H is closed under the operation, (2) $e \in H$, and (3) $a \in H \Rightarrow a^{-1} \in H$.

Proof. If $H \leq G$, (1) holds by definition. The identity e_H of H satisfies $e_H e_H = e_H = e_H e$ in G, so $e_H = e$ by cancellation, giving (2); the inverse of a in H is an inverse in G, hence equals a^{-1}, giving (3). Conversely (1)–(3) give $\mathcal{G}_2$ and $\mathcal{G}_3$ in H, and associativity is inherited from G. ■

For finite H, closure alone suffices (Exercise in the book: the powers of a repeat, so a^{-1} is a power of a). Equivalently, a nonempty H is a subgroup iff $ab^{-1} \in H$ for all $a, b \in H$.

Theorem 1.9 (5.17)

For $a \in G$, $H = \{a^n : n \in \mathbb{Z}\}$ is a subgroup of G, and it is the smallest subgroup containing a.

Proof. $a^r a^s = a^{r+s}$, $e = a^0$, $(a^r)^{-1} = a^{-r}$; apply Theorem 1.8. Any subgroup containing a contains all a^n by closure under products and inverses. ■

$H = \langle a \rangle$ is the **cyclic subgroup generated by** a; G is **cyclic** if $G = \langle a \rangle$ for some a, a **generator**. The **order** of a is $|\langle a \rangle|$.

1.5 Cyclic groups (§6)

Motivation. Cyclic groups are the simplest groups, generated by one element, and they can be classified completely: every cyclic group is $\mathbb{Z}$ or some $\mathbb{Z}_n$, and all their subgroups are cyclic again. They serve as the building blocks for the abelian groups of §11 and as the testing ground for later ideas such as order and index.

Theorem 1.10 (6.1)

Every cyclic group is abelian.

Proof. $a^r a^s = a^{r+s} = a^s a^r$. ■

Theorem 1.11 (6.3, division algorithm for $\mathbb{Z}$)

If $m \in \mathbb{Z}^+$ and $n \in \mathbb{Z}$, there are unique $q, r \in \mathbb{Z}$ with $n = mq + r$ and $0 \le r < m$.

Proof. The set $\{n - mq : q \in \mathbb{Z}\} \cap \{0, 1, 2, \dots\}$ is nonempty (take $q = -|n|$), so has a least element $r = n - mq$. If $r \ge m$, then $r - m = n - m(q+1) \ge 0$ is smaller.

Uniqueness: if $mq + r = mq' + r'$ with $0 \le r, r' < m$, then m divides $r - r'$ and $|r - r'| < m$, so $r = r'$ and $q = q'$. ■

Theorem 1.12 (6.6)

A subgroup of a cyclic group is cyclic.

Proof. Let $G = \langle a \rangle$ and $H \le G$. If $H = \{e\}$, $H = \langle e \rangle$. Otherwise H contains some a^n with $n > 0$ (if $a^{-n} \in H$, so is a^n); let m be the least such. For $b = a^n \in H$ write $n = mq + r$, $0 \le r < m$; then $a^r = a^n (a^m)^{-q} \in H$, so $r = 0$ by minimality, and $b = (a^m)^q$. Thus $H = \langle a^m \rangle$. ■

Corollary 1.13 (6.7)

The subgroups of $\langle \mathbb{Z}, + \rangle$ are exactly the groups $n\mathbb{Z}$, $n \in \mathbb{Z}$.

Proof. $\mathbb{Z} = \langle 1 \rangle$; a subgroup generated by n is $n\mathbb{Z}$. ■

For $r, s \in \mathbb{Z}^+$ the subgroup $\{nr + ms : n, m \in \mathbb{Z}\}$ is $d\mathbb{Z}$ for a unique $d > 0$, the **greatest common divisor** $\gcd(r, s)$: d divides r and s, every common divisor divides $d = nr + ms$, and d is a linear combination of r and s. r, s are **relatively prime** if $\gcd(r, s) = 1$. If r, s are relatively prime and r divides sm, then r divides m (multiply $1 = ar + bs$ by m).

Theorem 1.14 (6.10)

Let $G = \langle a \rangle$. If G is infinite, $G \cong \langle \mathbb{Z}, + \rangle$; if $|G| = n$, $G \cong \langle \mathbb{Z}_n, +_n \rangle$.

Proof. If all powers of a are distinct ($a^h = a^k$, $h > k$ would give $a^{h-k} = e$), then $a^m \mapsto m$ is a well-defined bijection $G \to \mathbb{Z}$, and $a^i a^j = a^{i+j}$ shows it is an isomorphism. Otherwise let n be the least positive integer with $a^n = e$. Then $e, a, \ldots, a^{n-1}$ are distinct (else a smaller positive power is e), and for any $m = nq + r$, $a^m = a^r$ (Theorem 1.11); so $G = \{a^0, \ldots, a^{n-1}\}$, $|G| = n$, and $a^i \mapsto i$ is an isomorphism with $\mathbb{Z}_n$, since $a^i a^j = a^{i+_n j}$. ■

In particular, if a has finite order n, then $a^m = e$ iff n divides m.

Theorem 1.15 (6.14)

Let $G = \langle a \rangle$ have order n and $b = a^s$. Then b generates a cyclic subgroup of order n/d, $d = \gcd(n, s)$. Also $\langle a^s \rangle = \langle a^t \rangle$ iff $\gcd(s, n) = \gcd(t, n)$.

Proof. $|\langle b \rangle|$ is the least $m > 0$ with $a^{sm} = e$, i.e. with $n \mid sm$. Write $n = dn'$, $s = ds'$ with $\gcd(n', s') = 1$; then $n \mid sm$ iff $n' \mid s'm$ iff $n' \mid m$. So the order is $n' = n/d$. For the last statement: in the cyclic group $\mathbb{Z}_n$, $\langle s \rangle$ is the unique subgroup of order n/d (it is $\langle d \rangle$: $d \in \langle s \rangle$ as $d = xs + yn$, and $\langle d \rangle$ has order n/d); so $\langle a^s \rangle = \langle a^{\gcd(s,n)} \rangle$, and distinct divisors d of n give subgroups of distinct orders. ■

Corollary 1.16 (6.16)

If a generates a cyclic group of order n, the other generators are the a^r with r relatively prime to n.

Proof. a^r generates iff its order $n/\gcd(r, n)$ is n. ■

Consequently a cyclic group of order n has exactly one subgroup of each order d dividing n, and $\phi(n)$ generators.

1.6 Generating sets and Cayley digraphs (§7)

Motivation. Few groups are cyclic, but most groups we meet are generated by a handful of elements. Describing a group by generators is economical, and the Cayley digraph turns the multiplication table into a picture in which a group element is a path.

Theorem 1.17 (7.4)

The intersection of any family of subgroups H_i ($i \in I$) of G is a subgroup of G.

Proof. If $a, b \in \bigcap H_i$, then ab and a^{-1} lie in every H_i, as does e; apply Theorem 1.8. ■

For $\{a_i : i \in I\} \subset G$, the intersection of all subgroups containing every a_i is the smallest such subgroup, the **subgroup generated by** $\{a_i\}$; if it is G, the a_i **generate** G.

Theorem 1.18 (7.6)

The subgroup H generated by $\{a_i : i \in I\}$ consists exactly of the finite products of integral powers of the a_i (a given a_i may occur several times).

Proof. Let K be the set of such finite products. K is closed under multiplication (juxtapose), contains $e = a_i^0$, and contains inverses: $(a_{i_1}^{n_1} \cdots a_{i_k}^{n_k})^{-1} = a_{i_k}^{-n_k} \cdots a_{i_1}^{-n_1}$. So $K \leq G$ contains each a_i, whence $H \subset K$. Conversely any subgroup containing the a_i contains all such products, so $K \subset H$. ■

A **Cayley digraph** of G for a generating set S has the elements of G as vertices and an arc of type s from g to gs for each $s \in S$. It is connected; at each vertex exactly one arc of each type starts and one ends; and two different paths from a vertex g ending at the same vertex describe equal products in G.

Example 1.4

$V = \{e, a, b, c\}$ is generated by $\{a, b\}$ but is not cyclic. $\mathbb{Z}_6$ is generated by $\{2, 3\}$, as $1 = 3 - 2$. The dihedral-like group with digraph two concentric triangles joined by spokes (generators a of order 3 and b of order 2 with $ba = a^2b$) is S_3.

Chapter 2. Permutations, Cosets, and Direct Products

2.1 Groups of permutations (§8)

Motivation. Groups first arose as permutations of the roots of a polynomial, and Cayley's theorem says nothing is lost by thinking of groups that way: every group is a group of permutations. Permutation groups therefore give concrete models for all groups.

A **permutation** of a set A is a bijection $A \to A$; their product $\sigma\tau$ is the composition $\sigma \circ \tau$ (apply τ first).

Theorem 2.1 (8.5)

The permutations of a nonempty set A form a group S_A under composition.

Proof. A composite of bijections is a bijection, composition is associative (Theorem 1.2), the identity map is an identity, and the inverse function of a bijection is an inverse. ■

$S_n = S_{\{1,\ldots,n\}}$ is the **symmetric group**, of order $n!$; it is nonabelian for $n \geq 3$. The **dihedral group** D_n $(n \geq 3)$ is the group of symmetries of a regular n-gon, a subgroup of S_n of order $2n$; $D_3 = S_3$.

Lemma 2.2 (8.15)

If $\phi : G \to G'$ is one-to-one and $\phi(xy) = \phi(x)\phi(y)$ for all x, y, then $\phi[G]$ is a subgroup of G' and ϕ is an isomorphism of G with $\phi[G]$.

Proof. $\phi(x)\phi(y) = \phi(xy) \in \phi[G]$. From $\phi(e)\phi(e) = \phi(e) = e'\phi(e)$, cancellation gives $\phi(e) = e'$; then $\phi(a)\phi(a^{-1}) = \phi(e) = e'$ gives $\phi(a)^{-1} = \phi(a^{-1}) \in \phi[G]$. So $\phi[G] \leq G'$ (Theorem 1.8), and $\phi : G \to \phi[G]$ is a bijection preserving the operation. ■

Theorem 2.3 (8.16, Cayley's theorem)

Every group is isomorphic to a group of permutations.

Proof. For $x \in G$ let $\lambda_x : G \to G$, $\lambda_x(g) = xg$. It is one-to-one ($xg_1 = xg_2 \Rightarrow g_1 = g_2$) and onto ($\lambda_x(x^{-1}c) = c$), so $\lambda_x \in S_G$. The map $\phi(x) = \lambda_x$ is one-to-one ($\lambda_x(e) = x$) and $\lambda_{xy}(g) = xyg = \lambda_x(\lambda_y(g))$, so $\phi(xy) = \phi(x)\phi(y)$. Apply Lemma 2.2. ■

ϕ is the **left regular representation**; with $\rho_x(g) = gx^{-1}$ one gets the right regular representation.

2.2 Orbits, cycles, and the alternating groups (§9)

Motivation. To compute with permutations we need a normal form. Every permutation splits into disjoint cycles (its orbits), and every cycle into transpositions; the surprising fact that the parity of the number of transpositions is well defined produces the alternating group A_n, which will turn out to be simple for $n \geq 5$, the group-theoretic reason the quintic is not solvable by radicals.

For $\sigma \in S_A$, $a \sim b$ iff $b = \sigma^n(a)$ for some $n \in \mathbb{Z}$ is an equivalence relation; its cells are the **orbits** of σ. A **cycle** is a permutation with at most one orbit of more than one element; $(a_1, a_2, \ldots, a_m)$ sends $a_i \mapsto a_{i+1}$, $a_m \mapsto a_1$, its **length** is m; a cycle of length 2 is a **transposition**.

Theorem 2.4 (9.8)

Every permutation of a finite set is a product of disjoint cycles.

Proof. Let $B_1, \dots, B_r$ be the orbits of σ and μ_i the cycle that agrees with σ on B_i and is the identity elsewhere (on an orbit $B = \{b, \sigma(b), \dots, \sigma^{m-1}(b)\}$, σ acts as the cycle $(b, \sigma(b), \dots, \sigma^{m-1}(b))$). The orbits are disjoint, so $\sigma = \mu_1\mu_2 \cdots \mu_r$: each x lies in exactly one B_i, and only μ_i moves it. ■

Disjoint cycles commute, and the decomposition is unique up to order (the cycles are determined by the orbits).

Corollary 2.5 (9.12)

Any permutation of a finite set with at least two elements is a product of transpositions.

Proof. $(a_1, a_2, \dots, a_m) = (a_1, a_m)(a_1, a_{m-1}) \cdots (a_1, a_2)$, and the identity is $(1, 2)(1, 2)$. ■

Theorem 2.6 (9.15)

No permutation in S_n is both a product of an even number and of an odd number of transpositions.

Proof. (Orbit counting.) Let $\sigma \in S_n$ have r orbits (counting fixed points) and set $N(\sigma) = n - r$. For a transposition $\tau = (i, j)$, $N(\tau\sigma) = N(\sigma) \pm 1$: if i, j lie in the same orbit of σ, write the cycle as $(i, b_1, \dots, b_h, j, c_1, \dots, c_k)$; then (i, j) times it is $(i, c_1, \dots, c_k)(j, b_1, \dots, b_h)$, so the number of orbits increases by 1. If i, j lie in different orbits, the same computation read backwards shows the two orbits merge, decreasing the count by 1. Since $N(\mathrm{id}) = 0$, a product of k transpositions has $N \equiv k \pmod 2$. So the parity of k is determined by σ. ■

A permutation is **even** or **odd** according to this parity; a cycle of length m has parity of $m - 1$.

Theorem 2.7 (9.20)

For $n \geq 2$, the even permutations of $\{1, \dots, n\}$ form a subgroup A_n of S_n of order $n!/2$.

Proof. Products of even permutations are even, the identity is even, and the inverse of $\tau_1 \cdots \tau_k$ is $\tau_k \cdots \tau_1$; so $A_n \leq S_n$. Fix $\tau = (1, 2)$. $\lambda_\tau(\sigma) = \tau\sigma$ maps A_n into the odd permutations B_n and B_n into A_n, and $\lambda_\tau^2 = \mathrm{id}$; so it is a bijection $A_n \to B_n$ and $|A_n| = |B_n| = n!/2$. ■

A_n is the **alternating group**.

2.3 Cosets and the theorem of Lagrange (§10)

Motivation. How large can a subgroup be? Cosets of H chop G into equal-sized pieces, so $|H|$ must divide $|G|$. This single counting argument already forces strong conclusions: groups of prime order are cyclic, and $a^{|G|} = e$, which contains Fermat's little theorem.

Theorem 2.8 (10.1)

Let $H \leq G$. The relations $a \sim_L b \iff a^{-1}b \in H$ and $a \sim_R b \iff ab^{-1} \in H$ are equivalence relations on G.

Proof. For $\sim_L$: $a^{-1}a = e \in H$; if $a^{-1}b \in H$ then $(a^{-1}b)^{-1} = b^{-1}a \in H$; if $a^{-1}b, b^{-1}c \in H$ then their product $a^{-1}c \in H$. $\sim_R$ is analogous. ■

The cell of a under $\sim_L$ is the **left coset** $aH = \{ah : h \in H\}$ (as $a^{-1}b = h \iff b = ah$); under $\sim_R$ it is the **right coset** Ha. For abelian G they coincide.

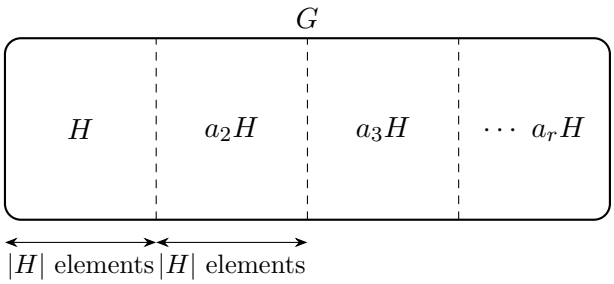


Figure. *The left cosets of H partition G into $r = (G : H)$ blocks of equal size $|H|$, so $|G| = r\,|H|$.*

Theorem 2.9 (10.10, Lagrange)

If H is a subgroup of a finite group G, then $|H|$ divides $|G|$.

Proof. $h \mapsto ah$ is a bijection $H \to aH$ (one-to-one by cancellation), so every left coset has $|H|$ elements. The left cosets partition G (Theorems 2.8, 1.1); if there are r of them, $|G| = r|H|$. ■

Corollary 2.10 (10.11)

Every group of prime order is cyclic.

Proof. For $a \neq e$, $|\langle a\rangle| > 1$ divides p, so equals p and $\langle a\rangle = G$. ■

Theorem 2.11 (10.12)

The order of an element of a finite group divides the order of the group.

Proof. It is the order of the cyclic subgroup it generates. ■

The number of left cosets of H in G is the **index** $(G : H)$; the map $aH \mapsto Ha^{-1}$ is a bijection between left and right cosets, so the index is the same for both.

Theorem 2.12 (10.14)

If $K \leq H \leq G$ with $(G : H)$ and $(H : K)$ finite, then $(G : K)$ is finite and $(G : K) = (G : H)(H : K)$.

Proof. Let $G = \bigsqcup_{i \leq r} a_iH$ and $H = \bigsqcup_{j \leq s} b_jK$. Then $a_iH = \bigsqcup_j a_ib_jK$, so $G = \bigcup_{i,j} a_ib_jK$. These rs cosets are distinct: if $a_ib_jK = a_pb_qK$, then since $b_jK, b_qK \subset H$, the cosets a_iH and a_pH meet, so $i = p$; then $b_jK = b_qK$ and $j = q$. ■

The converse of Lagrange fails: A_4 (order 12) has no subgroup of order 6.

2.4 Direct products and finitely generated abelian groups (§11)

Motivation. Direct products build new groups from old; the main theorem of the section goes backwards and says that every finitely generated abelian group is assembled in this way from

cyclic groups. Abelian groups are therefore completely classified by a list of integers, which lets us enumerate, for instance, all abelian groups of order 360.

Theorem 2.13 (11.2)

For groups $G_1, \ldots, G_n$, $\prod G_i$ is a group (the **direct product**) under $(a_1, \ldots, a_n)(b_1, \ldots, b_n) = (a_1 b_1, \ldots, a_n b_n)$.

Proof. The operation is componentwise, so associativity holds componentwise; $(e_1, \ldots, e_n)$ is the identity and $(a_1^{-1}, \ldots, a_n^{-1})$ the inverse. ■

Theorem 2.14 (11.5)

$\mathbb{Z}_m \times \mathbb{Z}_n$ is cyclic, and isomorphic to $\mathbb{Z}_{mn}$, iff $\gcd(m, n) = 1$.

Proof. If $\gcd(m, n) = 1$, the order of $(1, 1)$ is the least k divisible by both m and n, namely mn, so $(1, 1)$ generates. If $d = \gcd(m, n) > 1$, then mn/d is divisible by m and n, so every element (r, s) satisfies $(mn/d)(r, s) = (0, 0)$; no element has order mn. ■

Corollary 2.15 (11.6)

$\prod_{i=1}^{n} \mathbb{Z}_{m_i}$ is cyclic and isomorphic to $\mathbb{Z}_{m_1 \cdots m_n}$ iff the m_i are pairwise relatively prime.

Proof. Induction using Theorem 2.14 and $\prod_{i \le n} \mathbb{Z}_{m_i} \cong (\prod_{i<n} \mathbb{Z}_{m_i}) \times \mathbb{Z}_{m_n}$: if pairwise relatively prime, $m_1 \cdots m_{n-1}$ and m_n are relatively prime. Conversely if $\gcd(m_i, m_j) > 1$, the subgroup $\mathbb{Z}_{m_i} \times \mathbb{Z}_{m_j}$ is not cyclic, but subgroups of cyclic groups are cyclic. ■

Theorem 2.16 (11.9)

If $a_i \in G_i$ has finite order r_i, then $(a_1, \ldots, a_n)$ has order $\operatorname{lcm}(r_1, \ldots, r_n)$.

Proof. $(a_1, \ldots, a_n)^k = (e_1, \ldots, e_n)$ iff $r_i \mid k$ for all i; the least such $k > 0$ is the lcm. ■

Theorem 2.17 (11.12, fundamental theorem of finitely generated abelian groups)

Every finitely generated abelian group is isomorphic to

$$\mathbb{Z}_{(p_1)^{r_1}} \times \mathbb{Z}_{(p_2)^{r_2}} \times \cdots \times \mathbb{Z}_{(p_n)^{r_n}} \times \mathbb{Z} \times \cdots \times \mathbb{Z},$$

with primes p_i (not necessarily distinct) and $r_i \in \mathbb{Z}^+$. The direct product is unique except for the order of the factors; the number of factors $\mathbb{Z}$ is the **Betti number**.

Proof [Supplement].

Step 1. (Stacked bases.) Write G additively, generated by $x_1, \ldots, x_n$. The map $\mathbb{Z}^n \to G$, $(m_i) \mapsto \sum m_i x_i$, is an onto homomorphism with kernel K, and $G \cong \mathbb{Z}^n / K$ (Theorem 3.9). We show: there are a basis $f_1, \ldots, f_n$ of $\mathbb{Z}^n$ and integers $d_1 \mid d_2 \mid \cdots \mid d_k$ ($k \le n$, $d_i > 0$) such that $d_1 f_1, \ldots, d_k f_k$ is a basis of K. Induction on n; if $K = 0$ take $k = 0$. Over all bases $f_1, \ldots, f_n$ of $\mathbb{Z}^n$ and all nonzero $v = \sum c_i f_i \in K$, choose one in which some coefficient has the least possible positive absolute value d; renumbering and changing signs, $c_1 = d$.

(i) d divides every c_j: if $c_j = qd + r$ with $0 < r < d$, the basis $f_1 + q f_j, f_2, \ldots, f_n$ (still a basis, as $f_1 = (f_1 + q f_j) - q f_j$) gives v the j-th coefficient $c_j - qd = r$, contradicting minimality. So $v = d g_1$ with $g_1 = f_1 + \sum_{i>1} (c_i/d) f_i$, and $g_1, f_2, \ldots, f_n$ is a basis.

(ii) For $w = ag_1 + \sum_{i>1} b_i f_i \in K$, d divides a (else $w - qv$ has coefficient $a - qd \in (0, d)$), and d divides every b_i (apply (i) to the element $v + (w - \frac{a}{d}v) = dg_1 + \sum b_i f_i$ of K). Hence $K = \mathbb{Z}\, dg_1 \oplus K'$ with $K' = K \cap \langle f_2, \dots, f_n \rangle$, and d divides all coefficients of elements of K'.

(iii) By induction, $\langle f_2, \dots, f_n \rangle \cong \mathbb{Z}^{n-1}$ has a basis $g_2, \dots, g_n$ with $d_2 g_2, \dots, d_k g_k$ a basis of K' and $d_2 \mid \cdots \mid d_k$. The coordinates of g_2 with respect to $f_2, \dots, f_n$ have gcd 1 (it is part of a basis), and d divides all coordinates of $d_2 g_2$, so $d \mid d_2$. Put $d_1 = d$.

Step 2. (Existence.) Then $G \cong \mathbb{Z}^n / K \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_k} \times \mathbb{Z}^{n-k}$, via $\sum m_i f_i \mapsto (m_1 \bmod d_1, \dots, m_k \bmod d_k, m_{k+1}, \dots, m_n)$, an onto homomorphism with kernel K. Drop factors with $d_i = 1$, and split each $\mathbb{Z}_d$ with $d = \prod p_j^{s_j}$ as $\prod \mathbb{Z}_{p_j^{s_j}}$ (Corollary 2.15).

Step 3. (Uniqueness.) The torsion subgroup T (elements of finite order) corresponds to the finite factors, and $G/T \cong \mathbb{Z}^b$, where b is the number of $\mathbb{Z}$ factors; b is determined since $\mathbb{Z}^b / 2\mathbb{Z}^b$ has 2^b elements. For the finite part and a prime p, let T_p be the elements of p-power order, the product of the factors $\mathbb{Z}_{p^r}$ with this p. If $T_p \cong \prod_j \mathbb{Z}_{p^{r_j}}$, then $p^{k-1}T_p / p^k T_p$ has order p^{N_k}, where N_k is the number of j with $r_j \geq k$. Since these groups depend only on G, the multiset $\{r_j\}$ is determined. ■

Example 2.1

Abelian groups of order $360 = 2^3 \cdot 3^2 \cdot 5$, up to isomorphism: $3 \cdot 2 \cdot 1 = 6$ of them (partitions of 3 for the 2-part, of 2 for the 3-part), e.g. $\mathbb{Z}_8 \times \mathbb{Z}_9 \times \mathbb{Z}_5$ and $\mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_2 \times \mathbb{Z}_3 \times \mathbb{Z}_3 \times \mathbb{Z}_5$.

A group is **decomposable** if it is isomorphic to a direct product of two proper nontrivial subgroups.

Theorem 2.18 (11.15)

The finite indecomposable abelian groups are exactly the cyclic groups of prime power order.

Proof. By Theorem 2.17 a finite indecomposable abelian group has one factor $\mathbb{Z}_{p^r}$. Conversely, if $\mathbb{Z}_{p^r} \cong H \times K$ with nontrivial H, K of orders p^s, p^t $(s, t < r)$, then every element has order at most $p^{\max(s,t)} < p^r$, contradicting cyclicity. ■

Theorem 2.19 (11.16)

If m divides the order of a finite abelian group G, then G has a subgroup of order m.

Proof. Write $G \cong \prod \mathbb{Z}_{(p_i)^{r_i}}$ and $m = \prod_p p^{t_p}$ with $t_p \leq \sum_{p_i = p} r_i$. For each prime p choose $s_i \leq r_i$ (for $p_i = p$) with $\sum s_i = t_p$. The subgroup $\langle (p_i)^{r_i - s_i} \rangle \leq \mathbb{Z}_{(p_i)^{r_i}}$ has order $(p_i)^{s_i}$, and the product of these subgroups has order m. ■

Theorem 2.20 (11.17)

If m is square free, every abelian group of order m is cyclic.

Proof. In Theorem 2.17 the primes p_i must be distinct with $r_i = 1$; then $\prod \mathbb{Z}_{p_i} \cong \mathbb{Z}_m$ by Corollary 2.15. ■

2.5 Plane isometries (§12)

Motivation. The symmetries of the plane form a group, and classifying them gives a geometric illustration of the abstract theory: each isometry is a translation, rotation, reflection or glide reflection, and the possible symmetry groups of plane patterns are severely restricted.

An **isometry** of $\mathbb{R}^2$ is a distance-preserving map. Every isometry is a translation, a rotation, a reflection, or a glide reflection; it is affine, $\phi(x) = Ax + c$ with A orthogonal (an isometry fixing 0 preserves norms, hence inner products by polarization, hence is linear). Orientation-preserving isometries ($\det A = 1$) are translations and rotations.

Theorem 2.21 (12.5)

Every finite group G of isometries of the plane is isomorphic to $\mathbb{Z}_n$ or to a dihedral group D_n.

Proof.

Step 1. Let $G = \{\phi_1, \ldots, \phi_m\}$ and $P = \frac{1}{m}\sum_i \phi_i(0)$. Since each ϕ is affine, $\phi(P) = \frac{1}{m}\sum_i \phi(\phi_i(0)) = \frac{1}{m}\sum_i \phi_i(0) = P$ ($\phi\phi_i$ runs over G). Translating, assume $P = 0$; then G consists of rotations about 0 and reflections in lines through 0.

Step 2. The rotations form a subgroup H (a product of two orientation-preserving isometries is one). If $H \neq \{\mathrm{id}\}$, let ρ be the rotation in H by the least positive angle θ. Every rotation in H, by angle $\alpha \in [0, 2\pi)$, satisfies $\alpha = k\theta + \beta$ with $0 \leq \beta < \theta$, and ρ^{-k} composed with it rotates by β, so $\beta = 0$; hence $H = \langle\rho\rangle \cong \mathbb{Z}_n$.

Step 3. If $G = H$ we are done. Otherwise let $\mu \in G$ be a reflection. For any reflection $\mu' \in G$, $\mu^{-1}\mu'$ preserves orientation, so lies in H; hence $G = H \cup \mu H$ has order $2n$. Since $\mu\rho\mu^{-1} = \rho^{-1}$, G is generated by ρ (order n) and μ (order 2) with this relation, which is the dihedral group D_n (for $n = 1$, $\mathbb{Z}_2$; for $n = 2$, the Klein group D_2). ■

Chapter 3. Homomorphisms and Factor Groups

3.1 Homomorphisms (§13)

Motivation. An isomorphism says two groups are the same; a homomorphism says one group is a "shadow" of another. For example, parity $S_n \to \{\pm 1\}$ forgets almost everything about a permutation but keeps its sign. The kernel measures exactly what is forgotten, and the results here show that the kernel controls the whole map.

A map $\phi : G \to G'$ of groups is a **homomorphism** if $\phi(ab) = \phi(a)\phi(b)$ for all a, b. For $A \subset G$, $\phi[A]$ is its image; for $B \subset G'$, $\phi^{-1}[B]$ is its inverse image.

Example 3.1

The trivial homomorphism $g \mapsto e'$; $\det : GL(n, \mathbb{R}) \to \mathbb{R}^*$; the sign $S_n \to \{\pm 1\}$; reduction $\mathbb{Z} \to \mathbb{Z}_n$, $m \mapsto r$ (remainder); evaluation $f \mapsto f(c)$ on additive groups of functions; projections $G_1 \times G_2 \to G_i$; differentiation on C^1 functions.

Theorem 3.1 (13.12)

Let $\phi : G \to G'$ be a homomorphism.
(1) $\phi(e) = e'$.
(2) $\phi(a^{-1}) = \phi(a)^{-1}$.
(3) If $H \leq G$, then $\phi[H] \leq G'$.
(4) If $K' \leq G'$, then $\phi^{-1}[K'] \leq G$.
In short, ϕ preserves the identity, inverses and subgroups.

Proof. (1) $\phi(e)\phi(e) = \phi(ee) = \phi(e) = e'\phi(e)$; cancel.

(2) $\phi(a)\phi(a^{-1}) = \phi(e) = e'$.

(3) $\phi(a)\phi(b) = \phi(ab)$, $e' = \phi(e)$ and $\phi(a)^{-1} = \phi(a^{-1})$ lie in $\phi[H]$.

(4) If $\phi(a), \phi(b) \in K'$, then $\phi(ab) = \phi(a)\phi(b) \in K'$ and $\phi(a^{-1}) = \phi(a)^{-1} \in K'$; and $\phi(e) = e' \in K'$. ■

The **kernel** is $\ker \phi = \phi^{-1}[\{e'\}]$, a subgroup by (4).

Theorem 3.2 (13.15)

Let $H = \ker \phi$ and $a \in G$. Then $\phi^{-1}[\{\phi(a)\}] = aH = Ha$. In particular left and right cosets of H coincide.

Proof. $\phi(x) = \phi(a)$ iff $\phi(a^{-1}x) = e'$ iff $a^{-1}x \in H$ iff $x \in aH$. Likewise $\phi(x) = \phi(a)$ iff $\phi(xa^{-1}) = e'$ iff $x \in Ha$. ■

Corollary 3.3 (13.18)

A homomorphism is one-to-one iff its kernel is $\{e\}$.

Proof. If $\ker \phi = \{e\}$, the fibre of $\phi(a)$ is $a\{e\} = \{a\}$. Conversely, if ϕ is one-to-one, only e maps to $e' = \phi(e)$. ■

So to show ϕ is an isomorphism one checks the homomorphism property, $\ker \phi = \{e\}$, and surjectivity. A subgroup H is **normal**, $H \trianglelefteq G$, if $gH = Hg$ for all $g \in G$.

Corollary 3.4 (13.20)

$\ker \phi$ is a normal subgroup of G.

Proof. Theorem 3.2. ■

3.2 Factor groups (§14)

Motivation. Arithmetic modulo n makes $\mathbb{Z}$ into the smaller group $\mathbb{Z}_n$ by declaring multiples of n to be zero. Factor groups do the same for any group: collapse a subgroup H to the identity. This works exactly when H is normal, and the fundamental homomorphism theorem shows that every homomorphic image arises this way.

Theorem 3.5 (14.1)

Let $\phi : G \to G'$ be a homomorphism with kernel H. The cosets of H form a group G/H under $(aH)(bH) = (ab)H$, and $\mu : G/H \to \phi[G]$, $\mu(aH) = \phi(a)$, is an isomorphism. The product is well defined: it does not depend on the representatives a, b.

Proof. By Theorem 3.2, $aH = \phi^{-1}[\{\phi(a)\}]$, so μ is a well-defined bijection between the cosets and $\phi[G]$. Define the product of cosets by transporting the product of $\phi[G]$: the coset $\mu^{-1}(\phi(a)\phi(b)) = \mu^{-1}(\phi(ab)) = (ab)H$. Since this depends only on $\phi(a)$, $\phi(b)$, i.e. on the cosets, it is well defined, and G/H is a group isomorphic to $\phi[G]$ via μ. ■

Theorem 3.6 (14.4)

For $H \leq G$, left coset multiplication $(aH)(bH) = (ab)H$ is well defined iff $aH = Ha$ for all a.

Proof. Suppose it is well defined; let $x = ah \in aH$. Using the representatives x and a^{-1}, $(xH)(a^{-1}H) = (xa^{-1})H$, while also $(aH)(a^{-1}H) = eH = H$. So $aha^{-1} \in H$, i.e. $ah = h'a \in Ha$, and $aH \subset Ha$. Symmetrically $Ha \subset aH$ (use a^{-1} in place of a).

Conversely, if $aH = Ha$ for all a, and $a_1 = ah_1$, $b_1 = bh_2$, then $a_1b_1 = ah_1bh_2$; write $h_1b = bh_3$, so $a_1b_1 = ab(h_3h_2) \in (ab)H$. ■

Corollary 3.7 (14.5)

If $H \trianglelefteq G$, the cosets of H form a group G/H, the **factor group**, under $(aH)(bH) = (ab)H$.

Proof. The operation is well defined; associativity is inherited, $H = eH$ is the identity and $a^{-1}H$ is the inverse of aH. ■

Theorem 3.8 (14.9)

If $H \trianglelefteq G$, then $\gamma : G \to G/H$, $\gamma(x) = xH$, is a homomorphism with kernel H.

Proof. $\gamma(xy) = (xy)H = (xH)(yH)$, and $xH = H$ iff $x \in H$. ■

So the normal subgroups are exactly the kernels of homomorphisms.

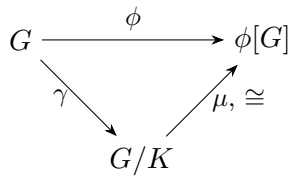


Figure. *The fundamental homomorphism theorem: $\phi = \mu \circ \gamma$, where $\gamma(g) = gK$ is the canonical map and $\mu(gK) = \phi(g)$ is an isomorphism.*

Theorem 3.9 (14.11, fundamental homomorphism theorem)

If $\phi : G \to G'$ is a homomorphism with kernel H, then $\phi[G]$ is a group and $\mu : G/H \to \phi[G]$, $\mu(gH) = \phi(g)$, is an isomorphism; with $\gamma(g) = gH$, $\phi = \mu \circ \gamma$.

Proof. Theorem 3.5, and $\mu(\gamma(g)) = \mu(gH) = \phi(g)$. ■

Example 3.2

$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$ via the reduction homomorphism. $\mathbb{R}/\mathbb{Z} \cong U$ via $x \mapsto e^{2\pi ix}$. $GL(n,\mathbb{R})/SL(n,\mathbb{R}) \cong \mathbb{R}^*$ via det.

Theorem 3.10 (14.13)

For $H \leq G$ the following are equivalent:
(1) $ghg^{-1} \in H$ for all $g \in G$, $h \in H$;
(2) $gHg^{-1} = H$ for all $g \in G$;
(3) $gH = Hg$ for all $g \in G$.

Proof. (1)⇒(2):

(1) gives $gHg^{-1} \subset H$; applying it to g^{-1}, $g^{-1}Hg \subset H$, i.e. $H \subset gHg^{-1}$.

(2)⇒(3): $gHg^{-1} = H$ gives $gH = Hg$ (multiply on the right by g).

(3)⇒(1): $gh \in gH = Hg$, so $gh = h'g$ and $ghg^{-1} = h' \in H$. ■

For fixed g, $i_g(x) = gxg^{-1}$ is an automorphism of G, the **inner automorphism** by g; condition (2) says H is invariant under all inner automorphisms.

3.3 Factor-group computations and simple groups (§15)

Motivation. Computing G/N is the main way to break a group into smaller pieces. Groups that cannot be broken further (no nontrivial proper normal subgroups) are the simple groups, the "primes" of group theory; here we see that A_n $(n \geq 5)$ is one of them.

Theorem 3.11 (15.8)

In $G = H \times K$, $\bar{H} = \{(h, e) : h \in H\}$ is normal and $G/\bar{H} \cong K$; similarly $G/\bar{K} \cong H$.

Proof. The projection $\pi_2(h,k) = k$ is an onto homomorphism with kernel $\bar{H}$; apply Theorem 3.9. ■

Theorem 3.12 (15.9)

A factor group of a cyclic group is cyclic.

Proof. If $G = \langle a \rangle$, then $(aN)^n = a^n N$, so $G/N = \langle aN \rangle$. ■

Example 3.3

$(\mathbb{Z}_4 \times \mathbb{Z}_6)/\langle (2,3) \rangle$ has order 12 and is generated by the images of $(1,0)$ (order 4) and $(0,1)$ (order 6, but $(0,3) \equiv (-2,0)$); it is isomorphic to $\mathbb{Z}_4 \times \mathbb{Z}_3 \cong \mathbb{Z}_{12}$. $(\mathbb{Z} \times \mathbb{Z})/\langle (1,1) \rangle \cong \mathbb{Z}$ via $(m,n) \mapsto m - n$. In S_3, A_3 is normal and $S_3/A_3 \cong \mathbb{Z}_2$.

A group is **simple** if it is nontrivial and has no proper nontrivial normal subgroups. Groups of prime order are simple.

Theorem 3.13 (15.15)

A_n is simple for $n \geq 5$.

Proof.

Step 1. A_n $(n \geq 3)$ is generated by 3-cycles: a product of two transpositions is a product of 3-cycles, since $(a,b)(c,d) = (a,c,b)(a,c,d)$ for distinct a,b,c,d and $(a,b)(a,c) = (a,c,b)$.

Step 2. If $N \trianglelefteq A_n$ contains a 3-cycle, then $N = A_n$: all 3-cycles are conjugate in A_n for $n \geq 5$. Indeed, given (a,b,c) and (a',b',c') choose $\sigma \in S_n$ with $\sigma(a) = a'$, $\sigma(b) = b'$, $\sigma(c) = c'$; then $\sigma(a,b,c)\sigma^{-1} = (a',b',c')$, and if σ is odd, replace it by $\sigma(d,e)$ with $d,e \notin \{a,b,c\}$, which gives the same conjugate.

Step 3. Let $N \neq \{\mathrm{id}\}$ be normal in A_n, $n \geq 5$; we find a 3-cycle in N. Let $\sigma \in N$, $\sigma \neq \mathrm{id}$, written as disjoint cycles.

Case 1: σ has a cycle of length $r \geq 4$, $\sigma = (a_1, \ldots, a_r)\mu$. With $\tau = (a_1, a_2, a_3)$, $N \ni \sigma^{-1}\tau\sigma\tau^{-1} = (a_1, a_3, a_r)$.

Case 2: σ has two 3-cycles, $\sigma = (a_1,a_2,a_3)(a_4,a_5,a_6)\mu$. With $\tau = (a_1,a_2,a_4)$, $\sigma^{-1}\tau\sigma\tau^{-1} = (a_1,a_4,a_2,a_6,a_3) \in N$, reducing to Case 1.

Case 3: $\sigma = (a_1,a_2,a_3)\mu$ with μ a product of disjoint transpositions; then $\sigma^2 = (a_1,a_3,a_2) \in N$.

Case 4: $\sigma = (a_1,a_2)(a_3,a_4)\mu$ with μ a product of disjoint transpositions. With $\tau = (a_1,a_2,a_3)$, $\alpha = \sigma^{-1}\tau\sigma\tau^{-1} = (a_1,a_3)(a_2,a_4) \in N$. Choose $i \notin \{a_1,\ldots,a_4\}$ (as $n \geq 5$) and $\beta = (a_1,a_3,i)$; then $\beta\alpha\beta^{-1} = (a_3,i)(a_2,a_4)$ and $\beta\alpha\beta^{-1}\alpha = (a_3,i)(a_1,a_3) = (a_1,i,a_3) \in N$.

In each case N contains a 3-cycle (the products are computed by following each symbol), so $N = A_n$ by Step 2. ■

Theorem 3.14 (15.16)

Let $\phi : G \to G'$ be a homomorphism. If $N \trianglelefteq G$, then $\phi[N] \trianglelefteq \phi[G]$. If $N' \trianglelefteq \phi[G]$ (or $N' \trianglelefteq G'$), then $\phi^{-1}[N'] \trianglelefteq G$.

Proof. For $\phi(g) \in \phi[G]$ and $\phi(n) \in \phi[N]$, $\phi(g)\phi(n)\phi(g)^{-1} = \phi(gng^{-1}) \in \phi[N]$. If $\phi(x) \in N'$ and $g \in G$, then $\phi(gxg^{-1}) = \phi(g)\phi(x)\phi(g)^{-1} \in N'$. ■

So normal subgroups of G/N correspond to normal subgroups of G containing N. A **maximal normal subgroup** of G is a proper normal subgroup contained in no larger proper normal subgroup.

Theorem 3.15 (15.18)

M is a maximal normal subgroup of G iff G/M is simple.

Proof. By Theorem 3.14 applied to $\gamma : G \to G/M$, a proper nontrivial normal subgroup of G/M has inverse image a normal subgroup strictly between M and G, and conversely a normal N with $M \subsetneq N \subsetneq G$ gives the proper nontrivial normal subgroup $\gamma[N]$ of G/M. ■

Theorem 3.16 (15.20)

The commutators $aba^{-1}b^{-1}$ generate a subgroup C of G (the **commutator subgroup**), C is normal, and for $N \trianglelefteq G$, G/N is abelian iff $C \leq N$.

Proof. C is a subgroup by definition (Theorem 1.18); the inverse of a commutator $aba^{-1}b^{-1}$ is the commutator $bab^{-1}a^{-1}$, so C consists of finite products of commutators. For $x \in C$ and $g \in G$, $gxg^{-1} = (gxg^{-1}x^{-1})x \in C$, so C is normal. G/N is abelian iff $abN = baN$ for all a, b iff $a^{-1}b^{-1}ab \in N$ for all a, b iff all commutators lie in N iff $C \leq N$. ■

3.4 Group action on a set (§16)

Motivation. Groups matter because they act: symmetries move points, permutations move letters, a group moves its own elements by conjugation. Formalizing actions gives the orbit–stabilizer theorem, a counting tool that will drive the Sylow theorems in Chapter 7.

An **action** of G on a set X is a map $G \times X \to X$, $(g, x) \mapsto gx$, with $ex = x$ and $(g_1g_2)x = g_1(g_2x)$; X is a **G-set**.

Theorem 3.17 (16.3)

For a G-set X and $g \in G$, $\sigma_g(x) = gx$ is a permutation of X, and $\phi : G \to S_X$, $\phi(g) = \sigma_g$, is a homomorphism with $\phi(g)(x) = gx$. Conversely, every homomorphism $G \to S_X$ defines an action by $gx = \phi(g)(x)$.

Proof. $\sigma_{g^{-1}}\sigma_g(x) = g^{-1}(gx) = ex = x$ and similarly $\sigma_g\sigma_{g^{-1}} = \mathrm{id}$, so σ_g is a bijection. $\sigma_{g_1g_2}(x) = (g_1g_2)x = g_1(g_2x) = \sigma_{g_1}\sigma_{g_2}(x)$. The converse is a direct check. ■

The action is **faithful** if only e acts as the identity, and **transitive** if for all x_1, x_2 some g has $gx_1 = x_2$. For $x \in X$, $G_x = \{g : gx = x\}$ is the **isotropy subgroup**, and $X_g = \{x : gx = x\}$.

Theorem 3.18 (16.12)

G_x is a subgroup of G for each $x \in X$.

Proof. $ex = x$; if $gx = hx = x$, then $(gh)x = g(hx) = x$ and $g^{-1}x = g^{-1}(gx) = x$. ■

Theorem 3.19 (16.14)

For $x_1, x_2 \in X$ let $x_1 \sim x_2$ iff $gx_1 = x_2$ for some $g \in G$. This is an equivalence relation on X.

Proof. $ex_1 = x_1$; if $gx_1 = x_2$ then $g^{-1}x_2 = x_1$; if $gx_1 = x_2$ and $hx_2 = x_3$ then $(hg)x_1 = x_3$. ■

The cells are the **orbits** $Gx = \{gx : g \in G\}$.

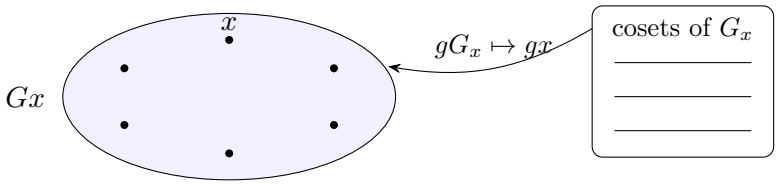


Figure. *Orbit–stabilizer: the points of the orbit Gx correspond one-to-one to the left cosets of the stabilizer G_x, so $|Gx| = (G : G_x)$.*

Theorem 3.20 (16.16, orbit–stabilizer)

$|Gx| = (G : G_x)$. If G is finite, $|Gx|$ divides $|G|$.

Proof. $gG_x \mapsto gx$ is well defined and one-to-one: $g_1x = g_2x$ iff $g_1^{-1}g_2 \in G_x$ iff $g_1G_x = g_2G_x$. It is onto Gx. Then use Lagrange. ■

Example 3.4

G acts on itself by left multiplication (transitive, faithful: Cayley) and by conjugation $g \cdot x = gxg^{-1}$, whose orbits are the conjugacy classes and whose isotropy groups are centralizers. S_n acts on $\{1, \ldots, n\}$; the rotation group of a cube acts on its 6 faces with isotropy groups of order 4, so it has order 24.

3.5 Applications of G-sets to counting (§17)

Motivation. How many essentially different ways are there to colour the faces of a cube? Burnside's formula answers such questions by counting orbits as the average number of fixed points.

Theorem 3.21 (17.1, Burnside's formula)

If G is finite, X a finite G-set and r the number of orbits, then $r \cdot |G| = \sum_{g \in G} |X_g|$.

Proof. Count $N = |\{(g, x) : gx = x\}|$ in two ways: $N = \sum_g |X_g| = \sum_x |G_x|$. By Theorem 3.20, $|G_x| = |G|/|Gx|$, so $\sum_{x \in \mathcal{O}} |G_x| = |G|$ for each orbit $\mathcal{O}$, and $N = r|G|$. ■

Corollary 3.22 (17.2)

The number of orbits is $\frac{1}{|G|} \sum_{g \in G} |X_g|$.

Example 3.5

Colouring the 6 faces of a cube with 6 different colours, up to rotation: only the identity fixes a colouring, so there are $6!/24 = 30$ distinguishable cubes. Necklaces of n beads with k colours up to rotation number $\frac{1}{n} \sum_{d|n} \phi(d) k^{n/d}$.

Chapter 4. Rings and Fields

4.1 Rings and fields (§18)

Motivation. $\mathbb{Z}$, $\mathbb{Q}$, $\mathbb{R}$, polynomials and matrices carry two operations linked by the distributive law. Rings are the common abstraction; fields are the rings in which division is possible, the setting of linear algebra and of solving polynomial equations.

Definition (Ring)

A **ring** $\langle R, +, \cdot \rangle$ is a set with two binary operations such that
$\mathcal{R}_1$: $\langle R, + \rangle$ is an abelian group;
$\mathcal{R}_2$: multiplication is associative;
$\mathcal{R}_3$: both distributive laws $a(b+c) = ab + ac$, $(a+b)c = ac + bc$ hold.
R is **commutative** if multiplication is; a **unity** is a multiplicative identity $1 \neq 0$; a **unit** is an element with a multiplicative inverse. A **division ring** (skew field) is a ring with unity in which every nonzero element is a unit; a **field** is a commutative division ring.

Theorem 4.1 (18.8)

In a ring, (1) $0a = a0 = 0$;
(2) $a(-b) = (-a)b = -(ab)$;
(3) $(-a)(-b) = ab$.

Proof. (1) $a0 + a0 = a(0+0) = a0 = a0 + 0$; cancel in the additive group. Similarly $0a = 0$.

(2) $ab + a(-b) = a(b + (-b)) = a0 = 0$, so $a(-b) = -(ab)$; likewise $(-a)b = -(ab)$.

(3) $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$. ■

A **ring homomorphism** $\phi : R \to R'$ satisfies $\phi(a+b) = \phi(a) + \phi(b)$ and $\phi(ab) = \phi(a)\phi(b)$; an **isomorphism** is a bijective homomorphism. Examples: $\mathbb{Z}_n$; $\mathbb{Z} \times \mathbb{Z}$; $M_n(R)$; the ring of functions $\mathbb{R} \to \mathbb{R}$; the reduction map $\mathbb{Z} \to \mathbb{Z}_n$. By the Chinese remainder argument, $\mathbb{Z}_{mn} \cong \mathbb{Z}_m \times \mathbb{Z}_n$ as rings when $\gcd(m,n) = 1$ ($k \mapsto (k \bmod m, k \bmod n)$ is a ring homomorphism, and an additive isomorphism by Theorem 2.14).

4.2 Integral domains (§19)

Motivation. In $\mathbb{Z}_6$ we have $2 \cdot 3 = 0$, so the familiar rule "$ab = 0$ implies $a = 0$ or $b = 0$" fails, and with it cancellation and the usual way of solving equations. Integral domains are exactly the rings where cancellation survives; every finite one is a field, and the characteristic records how the integer multiples of 1 behave.

Nonzero a, b with $ab = 0$ are **divisors of** 0.

Theorem 4.2 (19.3)

The divisors of 0 in $\mathbb{Z}_n$ are exactly the nonzero elements not relatively prime to n.

Proof. If $m \neq 0$ and $d = \gcd(m,n) \neq 1$, then $m \cdot (n/d) = (m/d)n = 0$ in $\mathbb{Z}_n$ with $n/d \neq 0$. If $\gcd(m,n) = 1$ and $ms = 0$ in $\mathbb{Z}_n$, then $n \mid ms$, so $n \mid s$ and $s = 0$. ■

Corollary 4.3 (19.4)

If p is prime, $\mathbb{Z}_p$ has no divisors of 0.

Theorem 4.4 (19.5)

The cancellation laws ($ab = ac$, $a \neq 0 \Rightarrow b = c$, and similarly on the right) hold in R iff R has no divisors of 0.

Proof. If cancellation holds and $ab = 0 = a0$ with $a \neq 0$, then $b = 0$. Conversely, if there are no divisors of 0 and $ab = ac$, $a \neq 0$, then $a(b - c) = 0$, so $b - c = 0$. ■

An **integral domain** is a commutative ring with unity and no divisors of 0.

Theorem 4.5 (19.9)

Every field is an integral domain.

Proof. If $ab = 0$ and $a \neq 0$, then $b = a^{-1}ab = 0$. ■

Theorem 4.6 (19.11)

Every finite integral domain is a field.

Proof. For $a \neq 0$, $x \mapsto ax$ is one-to-one (cancellation), hence onto the finite set D; so $ax = 1$ for some x. ■

Corollary 4.7 (19.12)

If p is prime, $\mathbb{Z}_p$ is a field.

The **characteristic** of R is the least $n \in \mathbb{Z}^+$ with $n \cdot a = 0$ for all $a \in R$, or 0 if there is none.

Theorem 4.8 (19.15)

Let R have unity. If $n \cdot 1 \neq 0$ for all $n \in \mathbb{Z}^+$, R has characteristic 0; otherwise the least n with $n \cdot 1 = 0$ is the characteristic.

Proof. If $n \cdot 1 = 0$, then $n \cdot a = (n \cdot 1)a = 0$ for all a; conversely $n \cdot a = 0$ for all a includes $a = 1$. ■

The characteristic of an integral domain is 0 or prime: if $n = rs$ with $1 < r, s < n$, then $(r \cdot 1)(s \cdot 1) = 0$.

4.3 Fermat's and Euler's theorems (§20)

Motivation. Lagrange's theorem, applied to the group of units of $\mathbb{Z}_n$, gives classical number theory almost for free: $a^{p-1} \equiv 1 \pmod{p}$ and $a^{\varphi(n)} \equiv 1 \pmod{n}$. The section also settles when the linear congruence $ax \equiv b \pmod{m}$ is solvable.

Theorem 4.9 (20.1, little theorem of Fermat)

If p is prime and $p \nmid a$, then $a^{p-1} \equiv 1 \pmod{p}$.

Proof. $\mathbb{Z}_p^*$ is a group of order $p-1$ (Corollary 4.7), so the order of $\bar{a}$ divides $p-1$ (Theorem 2.11). ■

Corollary 4.10 (20.2)

$a^p \equiv a \pmod{p}$ for every integer a and prime p.

Proof. Multiply Fermat by a; if $p \mid a$ both sides are 0. ■

Theorem 4.11 (20.6)

The nonzero elements of $\mathbb{Z}_n$ that are not divisors of 0 form a group G_n under multiplication modulo n.

Proof. They are the m with $\gcd(m,n)=1$ (Theorem 4.2); products of such are such; $1 \in G_n$; and for $a \in G_n$, $x \mapsto ax$ is one-to-one on the finite set G_n (cancellation), hence onto, so $ax=1$ for some $x \in G_n$. ■

$\phi(n) = |G_n|$ is **Euler's phi-function**.

Theorem 4.12 (20.8, Euler)

If $\gcd(a,n)=1$, then $a^{\phi(n)} \equiv 1 \pmod{n}$.

Proof. Lagrange in G_n. ■

Theorem 4.13 (20.10)

If $a \in \mathbb{Z}_m$ is relatively prime to m, then $ax=b$ has a unique solution in $\mathbb{Z}_m$ for each b.

Proof. a is a unit of $\mathbb{Z}_m$; $x = a^{-1}b$, unique by cancellation. ■

Corollary 4.14 (20.11)

If $\gcd(a,m)=1$, the solutions of $ax \equiv b \pmod{m}$ form exactly one residue class modulo m.

Theorem 4.15 (20.12)

Let $a, b \in \mathbb{Z}_m$ and $d = \gcd(a,m)$. Then $ax=b$ has a solution in $\mathbb{Z}_m$ iff $d \mid b$, and then it has exactly d solutions.

Proof. If $ax=b$ in $\mathbb{Z}_m$, then $ax-b=qm$ in $\mathbb{Z}$, so $d \mid b$. Conversely, if $d \mid b$, write $a=da_1$, $b=db_1$, $m=dm_1$. In $\mathbb{Z}$, $ax-b=qm$ iff $a_1x-b_1=qm_1$, so the solutions are the integers $x \equiv x_0 \pmod{m_1}$, where x_0 is the unique solution of $a_1x \equiv b_1 \pmod{m_1}$ (Theorem 4.13, as $\gcd(a_1,m_1)=1$). Modulo m these are $x_0, x_0+m_1, \ldots, x_0+(d-1)m_1$, which are d distinct classes. ■

Corollary 4.16 (20.13)

$ax \equiv b \pmod{m}$ is solvable iff $d=\gcd(a,m)$ divides b, and then the solutions form exactly d residue classes modulo m.

Example 4.1

$12x \equiv 27 \pmod{18}$: $d = 6 \nmid 27$, no solutions. $15x \equiv 27 \pmod{18}$: $d = 3$; $5x \equiv 9 \pmod 6$ gives $x \equiv 3 \pmod 6$, so $x \equiv 3, 9, 15 \pmod{18}$.

4.4 The field of quotients of an integral domain (§21)

Motivation. $\mathbb{Q}$ is built from $\mathbb{Z}$ by formal fractions. The same construction turns any integral domain D into the smallest field containing it; the lemmas below verify that the obvious definitions of $+$ and $\cdot$ on fractions are well defined.

Let D be an integral domain and $S = \{(a, b) \in D \times D : b \neq 0\}$; set $(a, b) \sim (c, d)$ iff $ad = bc$.

Lemma 4.17 (21.2)

$\sim$ is an equivalence relation on S.

Proof. Reflexive and symmetric by commutativity. If $ad = bc$ and $cf = de$, then $adf = bcf = bde$, so $d(af - be) = 0$; as $d \neq 0$, $af = be$. ∎

Let $F = S/\sim$ with classes $[(a, b)]$.

Lemma 4.18 (21.3)

$[(a, b)] + [(c, d)] = [(ad + bc, bd)]$ and $[(a, b)][(c, d)] = [(ac, bd)]$ are well-defined operations on F.

Proof. $bd \neq 0$. If $(a, b) \sim (a', b')$ and $(c, d) \sim (c', d')$, i.e. $ab' = a'b$, $cd' = c'd$, then $(ad + bc)b'd' = ab'dd' + bb'cd' = a'bdd' + bb'c'd = (a'd' + b'c')bd$, and $acb'd' = a'c'bd$. ∎

One checks that F is a field: $[(0, 1)]$ and $[(1, 1)]$ are the identities, $[(-a, b)]$ the additive inverse, and $[(b, a)]$ the inverse of $[(a, b)]$ for $a \neq 0$; the axioms reduce to identities in D.

Lemma 4.19 (21.4)

$i : D \to F$, $i(a) = [(a, 1)]$, is an isomorphism of D onto a subring of F.

Proof. $i(a) + i(b) = [(a + b, 1)]$, $i(a)i(b) = [(ab, 1)]$, and $[(a, 1)] = [(b, 1)]$ iff $a = b$. ∎

Theorem 4.20 (21.5)

Every integral domain D embeds in a field F in which every element is a quotient of two elements of D (a **field of quotients**).

Proof. Lemmas 4.17–4.19, with $[(a, b)] = i(a)i(b)^{-1}$. ∎

Theorem 4.21 (21.6)

If F is a field of quotients of D and L is a field containing D, then $\psi(a/b) = ab^{-1}$ (computed in L) gives an isomorphism of F onto a subfield of L with $\psi(a) = a$ for $a \in D$.

Proof. Well defined: $a/b = c/d$ means $ad = bc$, so $ab^{-1} = cd^{-1}$ in L. ψ preserves sums and products ($ab^{-1} + cd^{-1} = (ad + bc)(bd)^{-1}$, $ab^{-1}cd^{-1} = ac(bd)^{-1}$), and it is one-to-one because a field

homomorphism with $\psi(1) = 1$ has trivial kernel ($\psi(x) = 0$, $x \neq 0$, would give $1 = \psi(x)\psi(x^{-1}) = 0$). ■

Corollary 4.22 (21.8)

Every field containing an integral domain D contains a field of quotients of D.

Corollary 4.23 (21.9)

Any two fields of quotients of D are isomorphic (by an isomorphism fixing D).

Proof. If F, F' are fields of quotients, $\psi : F \to F'$ of Theorem 4.21 has image containing all ab^{-1}, i.e. all of F'. ■

4.5 Rings of polynomials (§22)

Motivation. To study equations $f(x) = 0$ algebraically we treat polynomials as formal objects and then "plug in" values. The evaluation homomorphism links the two views, so that finding a root of f becomes a question about the kernel of a ring homomorphism.

A **polynomial** with coefficients in R is $f(x) = \sum_{i \geq 0} a_i x^i$ with finitely many $a_i \neq 0$; polynomials are added and multiplied as usual, $(\sum a_i x^i)(\sum b_j x^j) = \sum_n (\sum_{i+j=n} a_i b_j) x^n$.

Theorem 4.24 (22.2)

$R[x]$ is a ring; it is commutative if R is, and has unity 1 if R does.

Proof. $\langle R[x], + \rangle$ is an abelian group (componentwise). Associativity: the coefficient of x^n in $(fg)h$ and in $f(gh)$ is $\sum_{i+j+k=n} a_i b_j c_k$. Distributivity follows from that of R. The rest is immediate. ■

$R[x_1, \ldots, x_n] = R[x_1, \ldots, x_{n-1}][x_n]$. If D is an integral domain, so is $D[x]$: the leading coefficient of fg is the product of the leading coefficients, and $\deg(fg) = \deg f + \deg g$.

Theorem 4.25 (22.4, evaluation homomorphism)

Let $F \leq E$ be fields and $\alpha \in E$. The map $\phi_\alpha : F[x] \to E$, $\phi_\alpha(\sum a_i x^i) = \sum a_i \alpha^i$, is a ring homomorphism with $\phi_\alpha(x) = \alpha$ and $\phi_\alpha(a) = a$ for $a \in F$.

Proof. Additivity is clear. For products, $\phi_\alpha(fg) = \sum_n (\sum_{i+j=n} a_i b_j) \alpha^n = (\sum a_i \alpha^i)(\sum b_j \alpha^j)$ since E is commutative. ■

α is a **zero** of f if $\phi_\alpha(f) = f(\alpha) = 0$.

Theorem 4.26 (22.11)

$x^2 - 2$ has no zero in $\mathbb{Q}$; thus $\sqrt{2}$ is irrational.

Proof. If $(m/n)^2 = 2$ in lowest terms, then $m^2 = 2n^2$, so m is even, $m = 2k$, $2k^2 = n^2$, and n is even, a contradiction. ■

4.6 Factorization of polynomials over a field (§23)

Motivation. Over a field, polynomials can be divided with remainder exactly like integers. This gives the factor theorem (roots correspond to linear factors) and makes irreducible polynomials the analogue of primes. Gauss's lemma and Eisenstein's criterion are the practical tools for proving irreducibility, which is what Chapter 6 will need to build field extensions.

Theorem 4.27 (23.1, division algorithm for $F[x]$)

Let $f, g \in F[x]$ with $g \neq 0$. There are unique $q, r \in F[x]$ with $f = gq + r$ and $r = 0$ or $\deg r < \deg g$.

Proof. Existence: the set $\{f - gs : s \in F[x]\}$ has an element r of least degree (or 0). If $\deg r = t \geq m = \deg g$, with leading coefficients c_t and b_m, then $r - (c_t/b_m)x^{t-m}g$ has smaller degree, a contradiction.

Uniqueness: if $gq + r = gq' + r'$, then $g(q - q') = r' - r$ has degree $< \deg g$ unless $q = q'$. ■

Corollary 4.28 (23.3, factor theorem)

$a \in F$ is a zero of f iff $x - a$ divides f.

Proof. $f = (x - a)q + r$ with $r \in F$; evaluating at a, $f(a) = r$. ■

Corollary 4.29 (23.5)

A nonzero $f \in F[x]$ of degree n has at most n zeros in F.

Proof. Induction: if a is a zero, $f = (x - a)q$ with $\deg q = n - 1$, and any other zero b satisfies $(b - a)q(b) = 0$, so $q(b) = 0$. ■

Corollary 4.30 (23.6)

A finite subgroup G of the multiplicative group F^* of a field is cyclic. In particular the multiplicative group of a finite field is cyclic.

Proof. By Theorem 2.17, $G \cong \mathbb{Z}_{d_1} \times \cdots \times \mathbb{Z}_{d_r}$ with prime powers d_i. Let $m = \operatorname{lcm}(d_i)$; then $a^m = 1$ for all $a \in G$, so $x^m - 1$ has $|G|$ zeros and $|G| \leq m$ (Corollary 4.29). But $m \leq \prod d_i = |G|$, with equality iff the d_i are pairwise relatively prime, in which case G is cyclic (Corollary 2.15). ■

A nonconstant $f \in F[x]$ is **irreducible over** F if it is not a product of two polynomials of lower degree in $F[x]$.

Theorem 4.31 (23.10)

A polynomial of degree 2 or 3 is reducible over F iff it has a zero in F.

Proof. A factorization into lower degrees must contain a linear factor $ax + b$, giving the zero $-b/a$; conversely use the factor theorem. ■

Theorem 4.32 (23.11, Gauss's lemma)

If $f \in \mathbb{Z}[x]$ factors in $\mathbb{Q}[x]$ into polynomials of degrees $r, s < \deg f$, it factors in $\mathbb{Z}[x]$ into polynomials of the same degrees.

Proof. Call $h \in \mathbb{Z}[x]$ primitive if its coefficients have gcd 1. The product of primitive polynomials is primitive: if a prime p divided all coefficients of gh, then reducing mod p gives $\bar{g}\bar{h} = 0$ in the integral domain $\mathbb{Z}_p[x]$, so $\bar{g} = 0$ or $\bar{h} = 0$. Now if $f = gh$ in $\mathbb{Q}[x]$, write $g = \frac{a}{b} g_1$, $h = \frac{c}{d} h_1$ with g_1, h_1 primitive in $\mathbb{Z}[x]$; then $bdf = ac\, g_1 h_1$. The gcd of the coefficients of the left side is a multiple of bd, and that of the right side is ac (as $g_1 h_1$ is primitive), so $bd \mid ac$ and $f = \frac{ac}{bd} g_1 \cdot h_1$ is a factorization in $\mathbb{Z}[x]$. ■

Corollary 4.33 (23.12)

If $f = x^n + a_{n-1}x^{n-1} + \cdots + a_0 \in \mathbb{Z}[x]$ with $a_0 \neq 0$ has a zero in $\mathbb{Q}$, it has a zero $m \in \mathbb{Z}$, and $m \mid a_0$.

Proof. A rational zero gives a linear factor in $\mathbb{Q}[x]$, hence by Theorem 4.32 a factorization $(x - m)(x^{n-1} + \cdots - a_0/m)$ in $\mathbb{Z}[x]$ (the leading coefficients multiply to 1), so $m \in \mathbb{Z}$ and $m \mid a_0$. ■

Theorem 4.34 (23.15, Eisenstein criterion)

Let p be prime and $f = a_n x^n + \cdots + a_0 \in \mathbb{Z}[x]$ with $a_n \not\equiv 0$, $a_i \equiv 0 \pmod{p}$ for $i < n$, and $a_0 \not\equiv 0 \pmod{p^2}$. Then f is irreducible over $\mathbb{Q}$.

Proof. Otherwise $f = (b_r x^r + \cdots + b_0)(c_s x^s + \cdots + c_0)$ in $\mathbb{Z}[x]$ with $r, s < n$. Since $p^2 \nmid a_0 = b_0 c_0$ but $p \mid a_0$, exactly one of b_0, c_0, say b_0, is divisible by p. As $p \nmid a_n = b_r c_s$, $p \nmid b_r$; let i be least with $p \nmid b_i$, $0 < i \leq r < n$. Then $a_i = b_i c_0 + b_{i-1} c_1 + \ldots$, where p divides every term except $b_i c_0$; so $p \nmid a_i$, a contradiction. ■

Corollary 4.35 (23.17)

The cyclotomic polynomial $\Phi_p(x) = \frac{x^p - 1}{x - 1} = x^{p-1} + \cdots + x + 1$ is irreducible over $\mathbb{Q}$ for every prime p.

Proof. $\Phi_p(x+1) = \frac{(x+1)^p - 1}{x} = x^{p-1} + \binom{p}{1} x^{p-2} + \cdots + \binom{p}{p-1}$, and $p \mid \binom{p}{k}$ for $0 < k < p$ while $\binom{p}{p-1} = p$; Eisenstein applies. A factorization of $\Phi_p(x)$ would give one of $\Phi_p(x+1)$. ■

Theorem 4.36 (23.18)

If $p(x)$ is irreducible in $F[x]$ and $p \mid rs$, then $p \mid r$ or $p \mid s$.

Proof. The set $\{ap + br\}$ is an ideal of $F[x]$, hence equals $\langle d \rangle$ (every ideal of $F[x]$ is principal, by the division algorithm as in Theorem 5.18). d divides p, so d is a unit or an associate of p. If $p \nmid r$, then d is a unit, say $d = 1 = ap + br$; then $s = aps + brs$ is divisible by p. ■

Corollary 4.37 (23.19)

If p is irreducible and divides $r_1 \cdots r_n$, then p divides some r_i.

Theorem 4.38 (23.20)

Every nonconstant $f \in F[x]$ is a product of irreducible polynomials, unique up to order and nonzero constant factors.

Proof. Existence by induction on degree: f is irreducible or a product of lower-degree factors.

Uniqueness: if $p_1 \cdots p_r = q_1 \cdots q_s$, then p_1 divides some q_j (Corollary 4.37), and as q_j is irreducible, $q_j = up_1$ with $u \in F^*$; cancel and induct. ■

4.7 Noncommutative examples (§24)

Motivation. Not every interesting ring is commutative: endomorphism rings, group rings and the quaternions are basic examples. The section ends with Wedderburn's theorem, which shows that noncommutativity cannot appear in finite division rings.

An **endomorphism** of an abelian group A is a homomorphism $A \to A$; $(\phi + \psi)(a) = \phi(a) + \psi(a)$.

Theorem 4.39 (24.1)

The endomorphisms $\mathrm{End}(A)$ of an abelian group form a ring under addition and composition.

Proof. $\phi + \psi$ is an endomorphism because A is abelian; $\langle \mathrm{End}(A), + \rangle$ is an abelian group with zero map and $-\phi$. Composition is associative, and $\phi(\psi + \chi) = \phi\psi + \phi\chi$ (as ϕ is additive) and $(\psi + \chi)\phi = \psi\phi + \chi\phi$ (by definition). ■

For a group G and commutative ring R with unity, the **group ring** RG consists of finite formal sums $\sum a_g g$ with $(\sum a_g g)(\sum b_h h) = \sum_k (\sum_{gh=k} a_g b_h)k$.

Theorem 4.40 (24.4)

RG is a ring.

Proof. Addition is componentwise. Associativity of multiplication: the coefficient of k in both triple products is $\sum_{ghl=k} a_g b_h c_l$, by associativity in G. Distributivity follows from that in R. ■

The **quaternions** $\mathbb{H}$ are $a_1 + a_2 i + a_3 j + a_4 k$ ($a_i \in \mathbb{R}$) with $i^2 = j^2 = k^2 = -1$, $ij = k = -ji$, $jk = i = -kj$, $ki = j = -ik$.

Theorem 4.41 (24.9)

$\mathbb{H}$ is a strictly skew field.

Proof. The map $a+bi+cj+dk \mapsto \begin{pmatrix} a+bi & c+di \\ -c+di & a-bi \end{pmatrix}$ is an additive injection into $M_2(\mathbb{C})$ preserving the multiplication rules of $1, i, j, k$; so $\mathbb{H}$ is isomorphic to a subring of $M_2(\mathbb{C})$, and associativity and distributivity hold. For $q = a_1+a_2 i+a_3 j+a_4 k \neq 0$ and $\bar{q} = a_1 - a_2 i - a_3 j - a_4 k$, $q\bar{q} = \bar{q}q = \sum a_i^2 > 0$, so $q^{-1} = \bar{q}/\sum a_i^2$. $ij \neq ji$. ■

Theorem 4.42 (24.10, Wedderburn)

Every finite division ring is a field.

Proof [Supplement]. (Witt.) Let D be a finite division ring and $Z = \{z : zx = xz\ \forall x\}$ its center, a finite field of q elements. D is a vector space over Z, so $|D| = q^n$. For $x \in D$ the centralizer $C(x) = \{y : yx = xy\}$ is a division ring containing Z, so $|C(x)| = q^d$; as D is a (left) vector space over $C(x)$, q^n is a power of q^d, so $d \mid n$, and $d < n$ if $x \notin Z$. The class equation of D^* (conjugation action, Theorem 3.20) gives

$$q^n - 1 = (q-1) + \sum \frac{q^n - 1}{q^{d_x} - 1},$$

summed over representatives of the noncentral classes, $d_x \mid n$, $d_x < n$. Let $\Phi_n(x) = \prod(x - \zeta)$ over the primitive n-th roots of unity ζ; from $x^n - 1 = \prod_{e|n} \Phi_e(x)$ and induction, $\Phi_n \in \mathbb{Z}[x]$ is monic, and $\Phi_n(x)$ divides $x^n - 1$ and $(x^n - 1)/(x^d - 1)$ in $\mathbb{Z}[x]$ for $d \mid n$, $d < n$. Hence the integer $\Phi_n(q)$ divides $q - 1$. But for $n > 1$ each $|q - \zeta| > q - 1 \geq 1$, so $|\Phi_n(q)| > q - 1$. Therefore $n = 1$ and $D = Z$ is commutative. ■

4.8 Ordered rings and fields (§25)

Motivation. Why can $\mathbb{C}$ not be ordered compatibly with its arithmetic? Ordered rings formalize the sign rules of $\mathbb{R}$; squares must be positive, which already excludes $i^2 = -1$.

A ring R is **ordered** if it has a subset P (positive elements) closed under addition and multiplication such that for each a exactly one of $a \in P$, $a = 0$, $-a \in P$ holds.

Theorem 4.43 (25.3)

In an ordered ring, squares of nonzero elements are positive, the characteristic is 0, and there are no divisors of 0.

Proof. If $a \in P$ then $a^2 \in P$; if $-a \in P$ then $a^2 = (-a)^2 \in P$. If $ab = 0$ with $a, b \neq 0$, then $\pm a, \pm b \in P$ give $\pm ab \in P$, contradicting $ab = 0$. If R has unity, $1 = 1^2 \in P$, so $n \cdot 1 \in P$ is never 0; in general $n \cdot a^2 \in P$ for $a \neq 0$. ■

Corollary 4.44 (25.4)

$\mathbb{Z}$ embeds in every ordered ring with unity, with its natural ordering induced, and the natural ordering is the only ordering of $\mathbb{Z}$. $\mathbb{C}$ cannot be ordered.

Proof. $n \mapsto n \cdot 1$ is one-to-one since the characteristic is 0, and $n \cdot 1 \in P$ for $n > 0$. In $\mathbb{Z}$, $1 = 1^2$ is positive, so all positive integers are. In $\mathbb{C}$, $-1 = i^2$ would be positive as well as 1. ■

Theorem 4.45 (25.5)

Defining $a < b$ iff $b - a \in P$: (trichotomy) exactly one of $a < b$, $a = b$, $b < a$ holds; (transitivity) $a < b$, $b < c \Rightarrow a < c$; (additivity) $a < b \Rightarrow a + c < b + c$; (multiplicativity) $a < b$, $0 < c \Rightarrow ac < bc$ and $ca < cb$.

Proof. Apply the axioms to $b-a$; $c-a = (c-b)+(b-a)$; $(b+c)-(a+c) = b-a$; $bc-ac = (b-a)c \in P$. ■

Theorem 4.46 (25.10)

If R is ordered by P and $\phi : R \to R'$ is a ring isomorphism, then $P' = \phi[P]$ orders R', and $\phi(a) < \phi(b)$ iff $a < b$.

Proof. ϕ preserves sums, products and negatives and is bijective, so P' satisfies the axioms; $\phi(b) - \phi(a) = \phi(b-a) \in P'$ iff $b - a \in P$. ■

Theorem 4.47 (25.13)

Let D be an ordered integral domain with positive set P and F its field of quotients. Then $P' = \{a/b : ab \in P\}$ is well defined, orders F, and induces the order of D; it is the only such order on F.

Proof. If $a/b = c/d$ $(ad = bc)$, then $ab\,d^2 = (ad)(bd) = (bc)(bd) = cd\,b^2$; since $b^2, d^2 \in P$, $ab \in P$ iff $cd \in P$ (multiplicativity and trichotomy). Closure: $\frac{a}{b} + \frac{c}{d} = \frac{ad+bc}{bd}$ with $(ad+bc)bd = ab\,d^2 + cd\,b^2 \in P$, and $\frac{a}{b}\frac{c}{d} = \frac{ac}{bd}$ with $acbd = (ab)(cd) \in P$. Trichotomy for a/b follows from that of ab. For $a \in D$, $a/1 \in P'$ iff $a \in P$.

Uniqueness: in any order of F extending that of D, $a/b = ab \cdot (1/b)^2$ is positive iff ab is. ■

So $\mathbb{Q}$ has only its natural ordering.

Chapter 5. Ideals and Factor Rings

5.1 Homomorphisms and factor rings (§26)

Motivation. This section repeats the group-theoretic story of Chapter 3 for rings: kernels of ring homomorphisms are ideals, factor rings R/N exist exactly for ideals N, and every homomorphic image is a factor ring. $\mathbb{Z}/n\mathbb{Z} = \mathbb{Z}_n$ is the model example.

Theorem 5.1 (26.3)

Let $\phi : R \to R'$ be a ring homomorphism.
(1) $\phi(0) = 0'$ and $\phi(-a) = -\phi(a)$.
(2) If S is a subring of R, $\phi[S]$ is a subring of R'; if S' is a subring of R', $\phi^{-1}[S']$ is a subring of R.
(3) If R has unity 1, then $\phi(1)$ is unity for $\phi[R]$ (if $\phi[R] \neq \{0'\}$).

Proof. (1) and the additive parts of (2) are Theorem 3.1 for $\langle R, + \rangle$. Closure under multiplication: $\phi(a)\phi(b) = \phi(ab)$, and $\phi(a), \phi(b) \in S'$ gives $\phi(ab) \in S'$.

(3) $\phi(1)\phi(a) = \phi(a) = \phi(a)\phi(1)$. ■

$\ker \phi = \phi^{-1}[\{0'\}]$.

Theorem 5.2 (26.5)

For $H = \ker \phi$ and $a \in R$, $\phi^{-1}[\{\phi(a)\}] = a + H = H + a$.

Proof. Theorem 3.2 for the additive groups. ■

Corollary 5.3 (26.6)

A ring homomorphism is one-to-one iff its kernel is $\{0\}$.

Theorem 5.4 (26.7)

If $\phi : R \to R'$ is a ring homomorphism with kernel H, the additive cosets of H form a ring R/H with $(a+H)+(b+H) = (a+b)+H$ and $(a+H)(b+H) = (ab)+H$, and $\mu(a+H) = \phi(a)$ is an isomorphism $R/H \to \phi[R]$.

Proof. As in Theorem 3.5: μ is a bijection from cosets onto $\phi[R]$, and the operations are transported from $\phi[R]$ ($\phi(a)\phi(b) = \phi(ab)$), so they are well defined and make R/H a ring isomorphic to $\phi[R]$. ■

Theorem 5.5 (26.9)

Let H be a subring of R. Multiplication of additive cosets $(a + H)(b + H) = ab + H$ is well defined iff $ah \in H$ and $hb \in H$ for all $a, b \in R$, $h \in H$.

Proof. If it is well defined, compare the representatives a, 0 and a, h of $a+H$, $0+H$: $a0+H = ah+H$, so $ah \in H$; similarly $hb \in H$. Conversely, if $a' = a+h_1$, $b' = b+h_2$, then $a'b' = ab+ah_2+h_1b+h_1h_2 \in ab + H$. ■

An additive subgroup N with $aN \subset N$ and $Nb \subset N$ for all $a, b \in R$ is an **ideal**.

Corollary 5.6 (26.14)

If N is an ideal of R, the additive cosets of N form a ring R/N, the **factor ring**, under $(a+N)+(b+N)=(a+b)+N$ and $(a+N)(b+N)=ab+N$.

Proof. The operations are well defined (Corollary 3.7 and Theorem 5.5); the ring axioms follow from those of R. ■

Theorem 5.7 (26.16)

If N is an ideal, $\gamma : R \to R/N$, $\gamma(x) = x+N$, is a ring homomorphism with kernel N.

Theorem 5.8 (26.17, fundamental homomorphism theorem)

If $\phi : R \to R'$ is a ring homomorphism with kernel N, then $\phi[R]$ is a ring and $\mu(x+N)=\phi(x)$ is an isomorphism $R/N \to \phi[R]$ with $\phi = \mu \circ \gamma$.

Proof. Theorem 5.4; the kernel is an ideal, since $\phi(ah)=\phi(a)0'=0'$ and $\phi(hb)=0'$. ■

Example 5.1

$\mathbb{Z}/n\mathbb{Z} \cong \mathbb{Z}_n$. The evaluation homomorphism $\phi_i : \mathbb{R}[x] \to \mathbb{C}$ has kernel $\langle x^2+1\rangle$ and image $\mathbb{C}$, so $\mathbb{R}[x]/\langle x^2+1\rangle \cong \mathbb{C}$.

5.2 Prime and maximal ideals (§27)

Motivation. The factor ring R/N inherits properties from N: it is a field exactly when N is maximal and an integral domain exactly when N is prime. This is the key to constructing fields: $F[x]/\langle p(x)\rangle$ is a field containing a root of p whenever p is irreducible, the starting point of Chapter 6.

Theorem 5.9 (27.5)

If R has unity and an ideal N contains a unit, then $N = R$.

Proof. If $u \in N$ is a unit, $1 = u^{-1}u \in N$, so $r = r1 \in N$ for all r. ■

Corollary 5.10 (27.6)

A field has no proper nontrivial ideals.

A **maximal ideal** is a proper ideal M not contained in any strictly larger proper ideal.

Theorem 5.11 (27.9)

Let R be commutative with unity. M is a maximal ideal iff R/M is a field.

Proof. ($\Rightarrow$) R/M is commutative with unity $1+M \neq 0+M$. For $a \notin M$, $N = \{ra+m : r \in R, m \in M\}$ is an ideal strictly containing M, so $N = R$ and $1 = ra+m$; then $(r+M)(a+M) = 1+M$.

($\Leftarrow$) If $M \subsetneq N$ with N an ideal, pick $a \in N - M$; $a+M$ has an inverse $b+M$, so $1-ab \in M \subset N$ and $1 = (1-ab)+ab \in N$, hence $N = R$. And $M \neq R$ since R/M is a field. ■

Corollary 5.12 (27.11)

A commutative ring with unity is a field iff it has no proper nontrivial ideals.

Proof. Corollary 5.10; conversely $\{0\}$ is maximal and $R \cong R/\{0\}$. ■

An ideal $N \neq R$ of a commutative ring is **prime** if $ab \in N$ implies $a \in N$ or $b \in N$.

Theorem 5.13 (27.15)

Let R be commutative with unity and $N \neq R$ an ideal. R/N is an integral domain iff N is prime.

Proof. $(a+N)(b+N) = N$ iff $ab \in N$, and $a+N = N$ iff $a \in N$; so "no divisors of 0 in R/N" is exactly primality. ■

Corollary 5.14 (27.16)

Every maximal ideal of a commutative ring with unity is prime.

Proof. A field is an integral domain. ■

Theorem 5.15 (27.17)

If R has unity, $\phi : \mathbb{Z} \to R$, $\phi(n) = n \cdot 1$, is a ring homomorphism.

Proof. $(n+m)\cdot 1 = n\cdot 1 + m\cdot 1$ and $(nm)\cdot 1 = (n\cdot 1)(m\cdot 1)$ by the distributive laws. ■

Corollary 5.16 (27.18)

If R has unity and characteristic $n > 1$, R contains a subring isomorphic to $\mathbb{Z}_n$; if characteristic 0, a subring isomorphic to $\mathbb{Z}$.

Proof. $\ker \phi = n\mathbb{Z}$; apply Theorem 5.8. ■

Theorem 5.17 (27.19)

A field F either has prime characteristic p and contains a subfield isomorphic to $\mathbb{Z}_p$, or has characteristic 0 and contains a subfield isomorphic to $\mathbb{Q}$.

Proof. The characteristic of an integral domain is 0 or prime. In the first case use Corollary 5.16. In the second, F contains a copy of $\mathbb{Z}$ and hence of its field of quotients $\mathbb{Q}$ (Corollary 4.22). ■

These subfields are the **prime fields**. An ideal $\langle a \rangle = \{ra : r \in R\}$ of a commutative ring with unity is **principal**.

Theorem 5.18 (27.24)

If F is a field, every ideal of $F[x]$ is principal.

Proof. Let $N \neq \{0\}$ and $g \in N$ nonzero of minimal degree. For $f \in N$, $f = gq + r$ with $r = 0$ or $\deg r < \deg g$; $r = f - gq \in N$, so $r = 0$. Hence $N = \langle g \rangle$. ■

Theorem 5.19 (27.25)

An ideal $\langle p(x)\rangle \neq \{0\}$ of $F[x]$ is maximal iff $p(x)$ is irreducible over F.

Proof. If $p = fg$ with lower degrees, then $\langle p\rangle \subsetneq \langle f\rangle \subsetneq F[x]$. Conversely, if p is irreducible and $\langle p\rangle \subset N = \langle g\rangle$, then $p = gq$, so g is a unit (then $N = F[x]$) or q is a unit (then $N = \langle p\rangle$). $\langle p\rangle \neq F[x]$ since p is not a unit. ■

So $F[x]/\langle p(x)\rangle$ is a field for irreducible p; for example $\mathbb{Z}_2[x]/\langle x^2 + x + 1\rangle$ is a field of 4 elements.

Theorem 5.20 (27.27)

If $p(x)$ is irreducible in $F[x]$ and divides $r(x)s(x)$, then it divides $r(x)$ or $s(x)$.

Proof. $\langle p\rangle$ is maximal, hence prime (Corollary 5.14), and $rs \in \langle p\rangle$. ■

5.3 Gröbner bases for ideals (§28)

Motivation. How do we decide whether a polynomial in several variables lies in a given ideal, or solve a system of polynomial equations? In one variable the division algorithm answers this; Gröbner bases are generating sets for which multivariable division behaves just as well. The Hilbert basis theorem guarantees that finite generating sets always exist.

For $S \subset F[x_1, \ldots, x_n]$ the **algebraic variety** $V(S) \subset F^n$ is the set of common zeros of the polynomials in S. A **basis** of an ideal I is a finite set $\{b_1, \ldots, b_r\} \subset I$ with $I = \langle b_1, \ldots, b_r\rangle = \{\sum f_i b_i\}$.

Theorem 5.21 (28.4)

The common zeros in F^n of $f_1, \ldots, f_r$ are the common zeros of all polynomials in $I = \langle f_1, \ldots, f_r\rangle$.

Proof. A common zero a of the f_i is a zero of every $\sum g_i f_i$, since evaluation at a is a ring homomorphism; the converse is clear as $f_i \in I$. ■

Theorem 5.22 (28.5, Hilbert basis theorem)

Every ideal of $F[x_1, \ldots, x_n]$ has a finite basis.

Proof [Supplement]. Call a commutative ring with unity *Noetherian* if all its ideals are finitely generated; fields are (Corollary 5.10). We show that R Noetherian implies $R[x]$ Noetherian; induction on n then gives the theorem. Suppose $I \subset R[x]$ is not finitely generated. Choose $f_1 \in I$ of least degree, and inductively $f_{k+1} \in I - \langle f_1, \ldots, f_k\rangle$ of least degree; then $d_k = \deg f_k$ is nondecreasing. Let a_k be the leading coefficient of f_k. The ideal $\langle a_1, a_2, \ldots\rangle$ of R is finitely generated, hence equals $\langle a_1, \ldots, a_N\rangle$ for some N; write $a_{N+1} = \sum_{i\le N} r_i a_i$. Then $g = f_{N+1} - \sum_{i\le N} r_i x^{d_{N+1}-d_i} f_i$ lies in $I - \langle f_1, \ldots, f_N\rangle$ and has degree $< d_{N+1}$, contradicting the choice of f_{N+1}. ■

A **power product** is $x_1^{m_1} \cdots x_n^{m_n}$. A **term order** $<$ is a total order of power products with $1 < P$ for $P \neq 1$ and $P < Q \Rightarrow PR < QR$; e.g. the lexicographic order with $x_1 > x_2 > \cdots > x_n$. It is a well-ordering (no infinite strictly decreasing sequence), by Dickson's lemma: every set of power products has finitely many minimal elements under divisibility (a consequence of Theorem 5.22 applied to the monomial ideal they generate). For $f \neq 0$ write $\mathrm{lp}(f)$ for its largest power product and $\mathrm{lt}(f)$ for its leading term. Division by $g_1, \ldots, g_r$ (repeatedly cancel the largest term divisible

by some $\operatorname{lp}(g_i)$) writes $f = \sum q_i g_i + r$ with $\operatorname{lp}(q_i g_i) \le \operatorname{lp}(f)$ and no term of r divisible by any $\operatorname{lp}(g_i)$; it terminates by the well-ordering.

Theorem 5.23 (28.6, property of the division algorithm)

If $f = gq + r$ in $F[\mathbf{x}]$, the common zeros of f, g are those of g, r, and the common divisors of f, g are those of g, r. Hence replacing f by r in a basis does not change the ideal.

Proof. $f \in \langle g, r\rangle$ and $r = f - gq \in \langle f, g\rangle$, so $\langle f, g\rangle = \langle g, r\rangle$; zeros and divisors follow. ■

$\{g_1, \dots, g_r\}$ is a **Gröbner basis** of $I = \langle g_1, \dots, g_r\rangle$ if for every nonzero $f \in I$ some $\operatorname{lp}(g_i)$ divides $\operatorname{lp}(f)$. For monic g_i, g_j with $L = \operatorname{lcm}(\operatorname{lp} g_i, \operatorname{lp} g_j)$, the **S-polynomial** is $S(g_i, g_j) = \frac{L}{\operatorname{lt} g_i} g_i - \frac{L}{\operatorname{lt} g_j} g_j$, in which the leading terms cancel.

Theorem 5.24 (28.12, Buchberger's criterion)

$G = \{g_1, \dots, g_r\}$ is a Gröbner basis of $\langle G\rangle$ iff every $S(g_i, g_j)$ reduces to 0 by repeated division by elements of G.

Proof [Supplement]. We may assume the g_i monic.

($\Rightarrow$) $S(g_i, g_j) \in I$; its remainder r on division lies in I and has no term divisible by any $\operatorname{lp}(g_k)$; if $r \neq 0$ this contradicts the Gröbner property for $\operatorname{lp}(r)$.

($\Leftarrow$) Reduction to 0 gives a representation $S(g_i, g_j) = \sum_k q_k g_k$ with $\operatorname{lp}(q_k g_k) \le \operatorname{lp} S(g_i, g_j) < L_{ij}$. Let $f \in I$ be nonzero. Among all representations $f = \sum h_i g_i$ choose one with $\delta = \max_i \operatorname{lp}(h_i g_i)$ minimal (well-ordering). Then $\operatorname{lp}(f) \le \delta$; if equality holds, $\operatorname{lp}(f) = \operatorname{lp}(h_i)\operatorname{lp}(g_i)$ for some i, and we are done. Suppose $\operatorname{lp}(f) < \delta$, and let T be the set of i with $\operatorname{lp}(h_i g_i) = \delta$, $\operatorname{lt}(h_i) = c_i P_i$. The terms in δ cancel, so $\sum_{i\in T} c_i = 0$. Put $p_i = P_i g_i$ (monic, $\operatorname{lp} = \delta$) and order $T = \{i_1, \dots, i_s\}$. Then

$$\sum_{i\in T} c_i p_i = \sum_{k<s} (c_{i_1} + \cdots + c_{i_k})(p_{i_k} - p_{i_{k+1}}),$$

and $p_{i_k} - p_{i_{k+1}} = \frac{\delta}{L} S(g_{i_k}, g_{i_{k+1}})$ with $L = L_{i_k i_{k+1}}$, which by hypothesis is $\sum_m q'_m g_m$ with all $\operatorname{lp}(q'_m g_m) < \delta$. Substituting into $f = \sum_{i\in T} c_i p_i + \sum_{i\in T}(h_i - c_i P_i) g_i + \sum_{i\notin T} h_i g_i$ gives a representation with maximum $< \delta$, a contradiction. ■

Buchberger's algorithm: add to G the nonzero remainders of the S-polynomials until all reduce to 0; it terminates because the ideal generated by the leading power products strictly grows and ascending chains of ideals stabilize (Theorem 5.22). Gaussian elimination and the Euclidean algorithm are the linear and one-variable cases.

Example 5.2

For $I = \langle xy, y^2 - y\rangle$ in $\mathbb{R}[x, y]$ (lex, $x > y$): $S(xy, y^2 - y) = y \cdot xy - x(y^2 - y) = xy$, which reduces to 0. So $\{xy, y^2 - y\}$ is a Gröbner basis, and $V(I) = \{(x, 0)\} \cup \{(0, 1)\}$.